

Seurat_tutorial

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Introduction

This RMarkdown contains a brief summary of the R package Seurat. Seurat is used for the analysis of single-cell RNA sequencing data, providing tools for data preprocessing, quality control, normalization, dimensionality reduction, clustering, differential expression, and cell-type annotation. Besides analyzing a dataset Seurat can also be used to create a variety of plots to visualize the data. The summary in this RMarkdown is based on the [Seurat guided clustering tutorial](#).

Purpose

The purpose of this RMarkdown is to understand the different functionalities of Seurat, and to learn how to use Seurat for analyzing the [GSE138852](#) count matrix.

Data

The data that is used in this RMarkdown can be accessed [here](#). This is a dataset of Peripheral Blood Mononuclear Cells (PBMC) freely available from 10X Genomics. This dataset is also used in the [Seurat guided clustering tutorial](#).

Packages

Prior to analysis several packages need to be loaded. The required packages are the following:

- `dplyr`: this package is used by Seurat for data manipulation.
- `Seurat`: this package needs to be loaded prior to running any Seurat specific functions.
- `patchwork`: the patchwork package can be used to combine several plots into a single figure.
- `here`: this package is used to make sure the file paths are reproducible.

Loading the dataset and creating a Seurat object

In order to perform analysis with Seurat, the data first needs to be loaded. This is done with `Read10X()` which is a Seurat function that can be used to load 10X genomics files such as matrices and .tsv files. After loading the dataset, the function `CreateSeuratObject()` is used to create a Seurat object that contains the raw data. In this object several filters can be applied or altered, which is shown in the following sections.

Pre-processing of the dataset

To make sure the data that will be analyzed is reliable it is important that low quality cells are removed from the dataset. In this case the quality doesn't refer to a Phred score (read quality), but to the amount of unique genes per cell, the percentage of mitochondrial genes and the total number of molecules per cell;

- The number of unique genes per cell are relevant since too few genes compared to the rest of the cells could mean that sequencing of this particular cell was insufficient or that the droplet did not contain a cell. If the number of genes is higher it could mean that two or more cells were captured in the same droplet.
- The number of counts says something about the number of different molecules or Unique Molecular Identifiers (UMIs) in one cell. If this number is too low it could mean that this is a poor capture and if this number is too high it could mean that there might be more than one cell in a droplet.
- The percentage of mitochondrial genes are a good indication about the general health of the cell, if this percentage is high it usually means that the cell is dying or stressed. Since it might not be representative for the condition that is analysed it is better to filter this out.

The first step in pre-processing is adding a column to the dataframe with the percentage of mitochondrial genes. To give a general idea of what the dataframe looks like after adding this column the first five rows are shown in the chunk below. In the first column the cell barcodes are shown, the second column contains the original identity of the sample the third, fourth and fifth column contain the number of the number of counts, unique genes and the percentage of mitochondrial genes.

##	orig.ident	nCount_RNA	nFeature_RNA	percent_mt
## AAACATACAACCAC-1	pbmc3k	2419	779	3.0177759
## AAACATTGAGCTAC-1	pbmc3k	4903	1352	3.7935958
## AAACATTGATCAGC-1	pbmc3k	3147	1129	0.8897363
## AAACCGTGCTTCCG-1	pbmc3k	2639	960	1.7430845
## AAACCGTGTATGCG-1	pbmc3k	980	521	1.2244898

The dataframe can also be visualised in a plot, this XXXXX

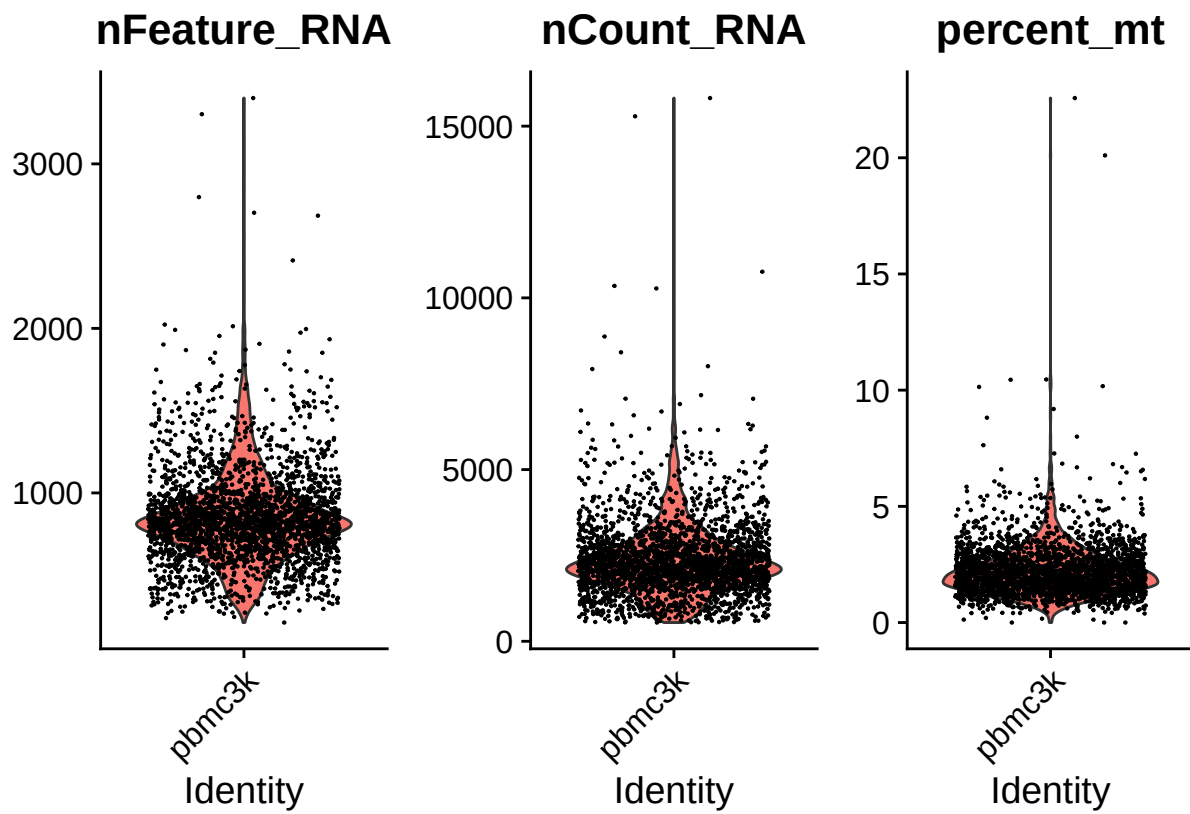


Figure 1: This plot shows the data of pbmc dataframe in three separate violin plots